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$$17. N = G \cos \theta = 43 \text{ N}, F_{\text{down}} = G \sin \theta = 12 \text{ N}$$

$$f_s = \mu_s N = 22 \text{ N} \quad f_k = \mu_k N = 15 \text{ N}$$

$$(a) F_{\text{down}} + P_1 = 17 \text{ N} < f_s = 22 \text{ N}, \text{ so } \vec{f}_1 = 17 \text{ N} \uparrow$$

$$(b) F_{\text{down}} + P_2 = 20 \text{ N} < f_s = 22 \text{ N}, \text{ so } \vec{f}_2 = 20 \text{ N} \uparrow$$

$$(c) F_{\text{down}} + P_3 = 27 \text{ N} > f_s = 22 \text{ N}, \text{ so } \vec{f}_3 = f_k = 15 \text{ N}$$

$$\text{so } \vec{f}_3 = 15 \text{ N} \uparrow$$

$$21. f = \mu Mg = 5.88 \text{ N}$$

$$(a) mg - f = (m + M) a \Rightarrow a = 0.807 \text{ m/s}^2$$

$$(b) mg - T = ma \Rightarrow T = 18.0 \text{ N}$$

35.

$$\mu_s N = mg \Rightarrow N = 480 \text{ N}$$

$$\begin{cases} F_{\text{min}} = (m + M) a \\ N = Ma \end{cases} \Rightarrow \begin{cases} a = 5.5 \text{ m/s}^2 \\ F_{\text{min}} = 570 \text{ N} \end{cases}$$

37.

$$(a) \frac{1}{2} C \rho A v_1^2 = \mu_k mg \Rightarrow v_1 = 90.0 \text{ m/s} = 324 \text{ km/h}$$

$$(b) v_2 = 2v_1 = 648 \text{ km/h}$$

(c) No. Because the speed is too fast.

$$42. \mu_s mg = m \frac{v^2}{r} \Rightarrow v = 13.7 \text{ m/s}$$

$$57. r = 25 \text{ cm} = 0.25 \text{ m}$$

$$Mg = m \frac{v^2}{r} \Rightarrow v = 2.02 \text{ m/s}$$